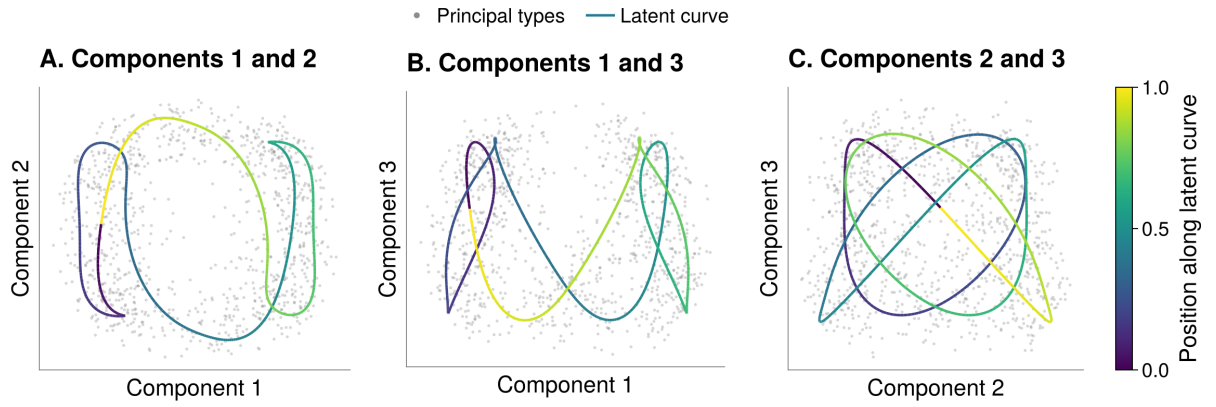
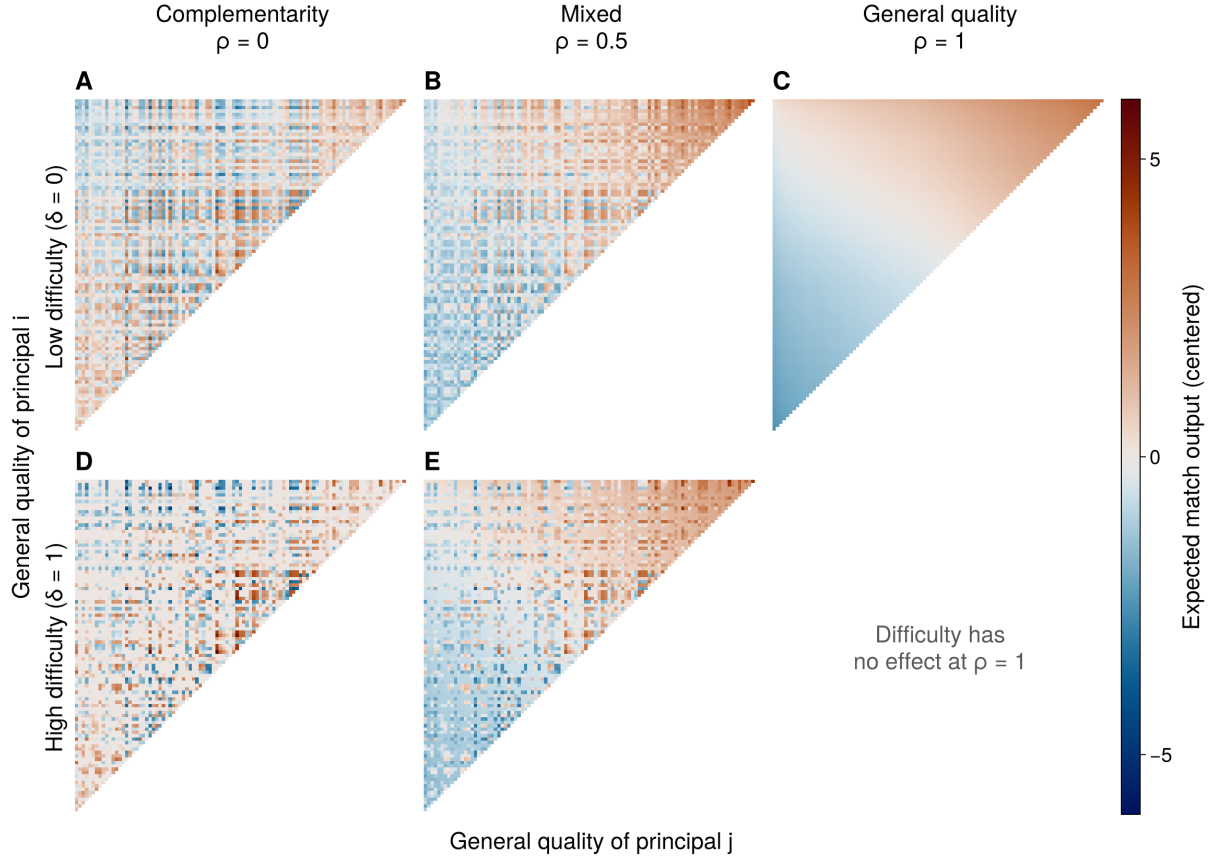


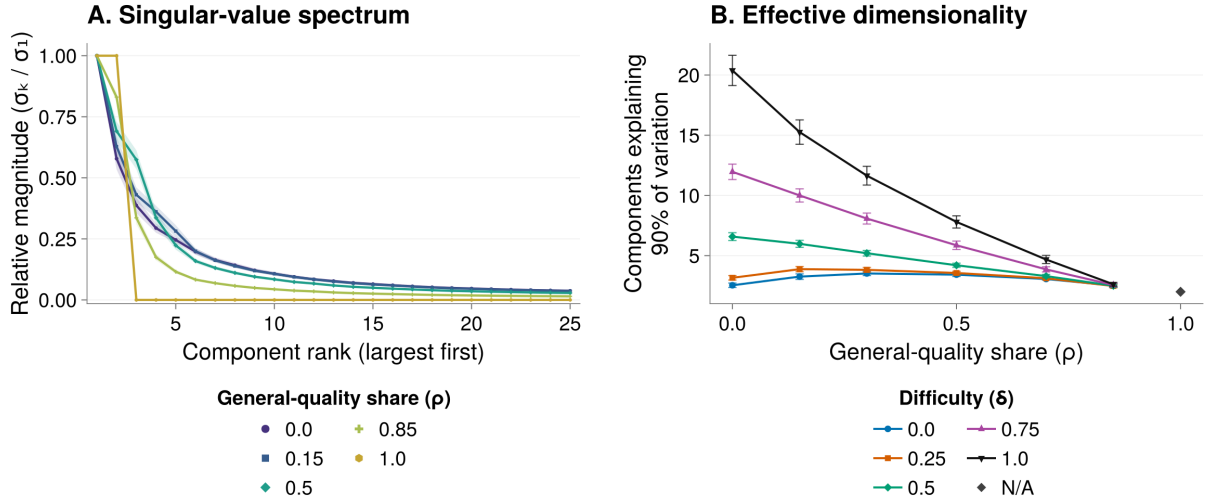
## Supplementary Material



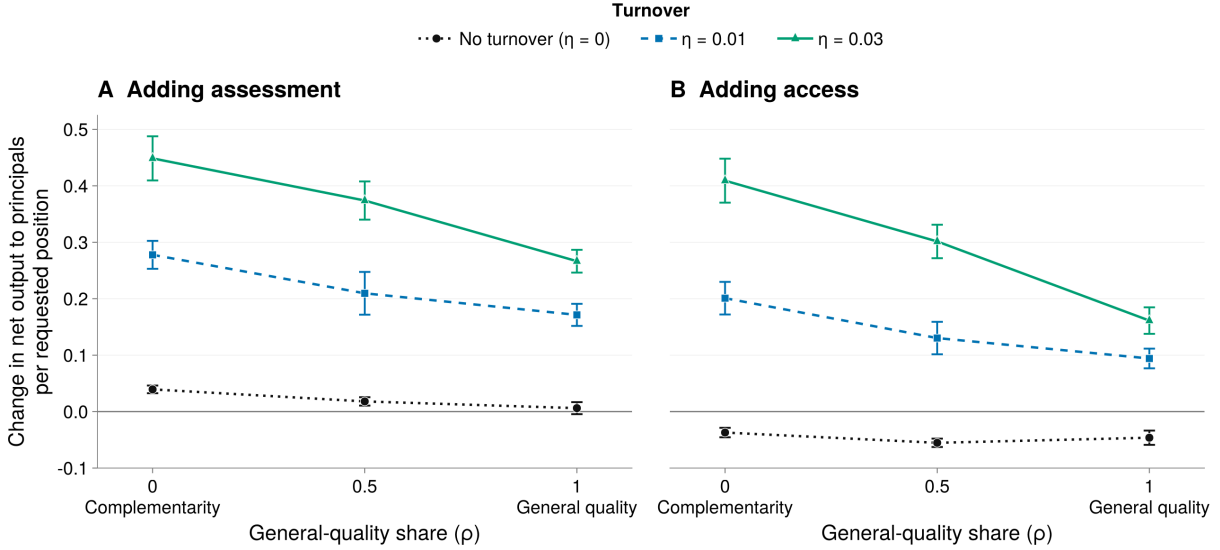
**Figure S1: Principal types and their latent curve.** All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.



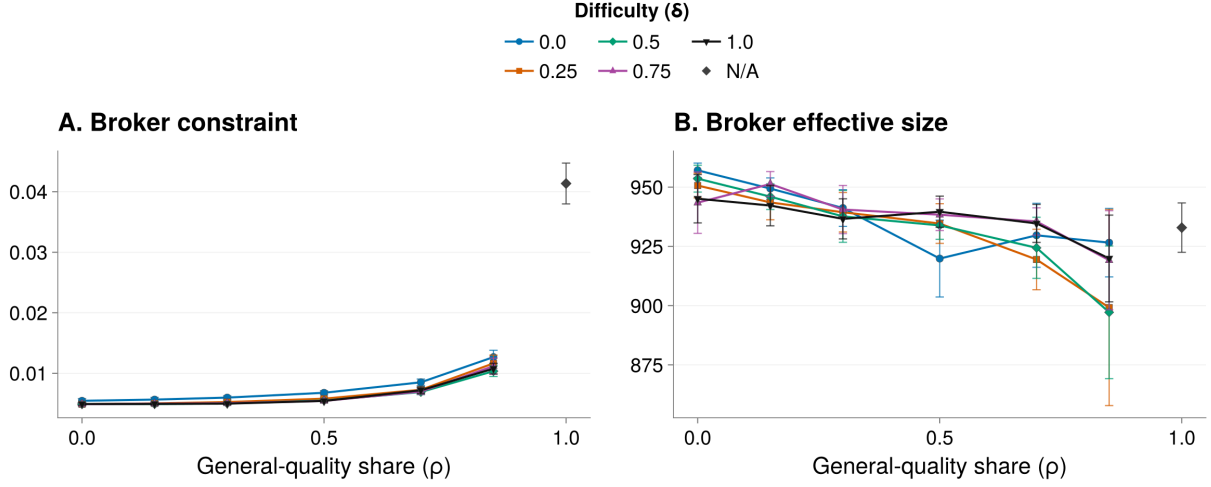
**Figure S2: Expected match output across pairs of principal types.** 100 types selected at evenly spaced general-quality ranks from realization 1 (1000 principals). Columns vary general-quality share  $\rho$ ; rows vary difficulty  $\delta$ . The  $\rho = 1$  condition appears once because difficulty is inactive. Types and matching-function objects  $c$ ,  $A$ , and  $B$  are fixed across panels. General quality increases rightward and upward. Only the upper triangle of each symmetric matrix is shown, excluding self-matches. A shared color scale shows expected output minus the grand mean, without match noise.



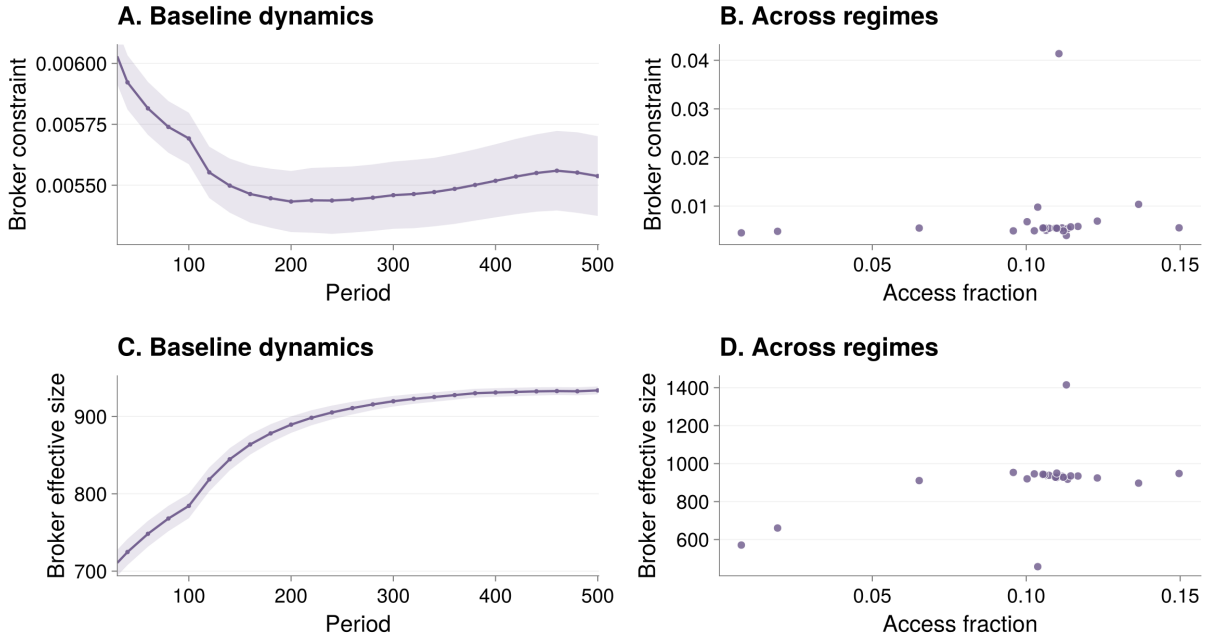
**Figure S3: Dimensionality of expected match output.** Centered conditional-output matrices, without match noise, summarized over 50 independent realizations. Types and matching-function objects  $c$ ,  $A$ , and  $B$  are fixed within each realization as  $\rho$  and  $\delta$  vary. A: the first 25 singular values at  $\delta = 0.5$ , divided by the largest within each realization. B: the fewest singular components accounting for 90% of the sum of squared singular values within each realization. Lines and markers are means; ribbons and whiskers are pointwise 95% Monte Carlo intervals. Colors and symbols identify general-quality share  $\rho$  in A and difficulty  $\delta$  in B. N/A denotes the unconnected gray diamond at  $\rho = 1$ : difficulty affects only complementarity and is not applicable under pure general quality.



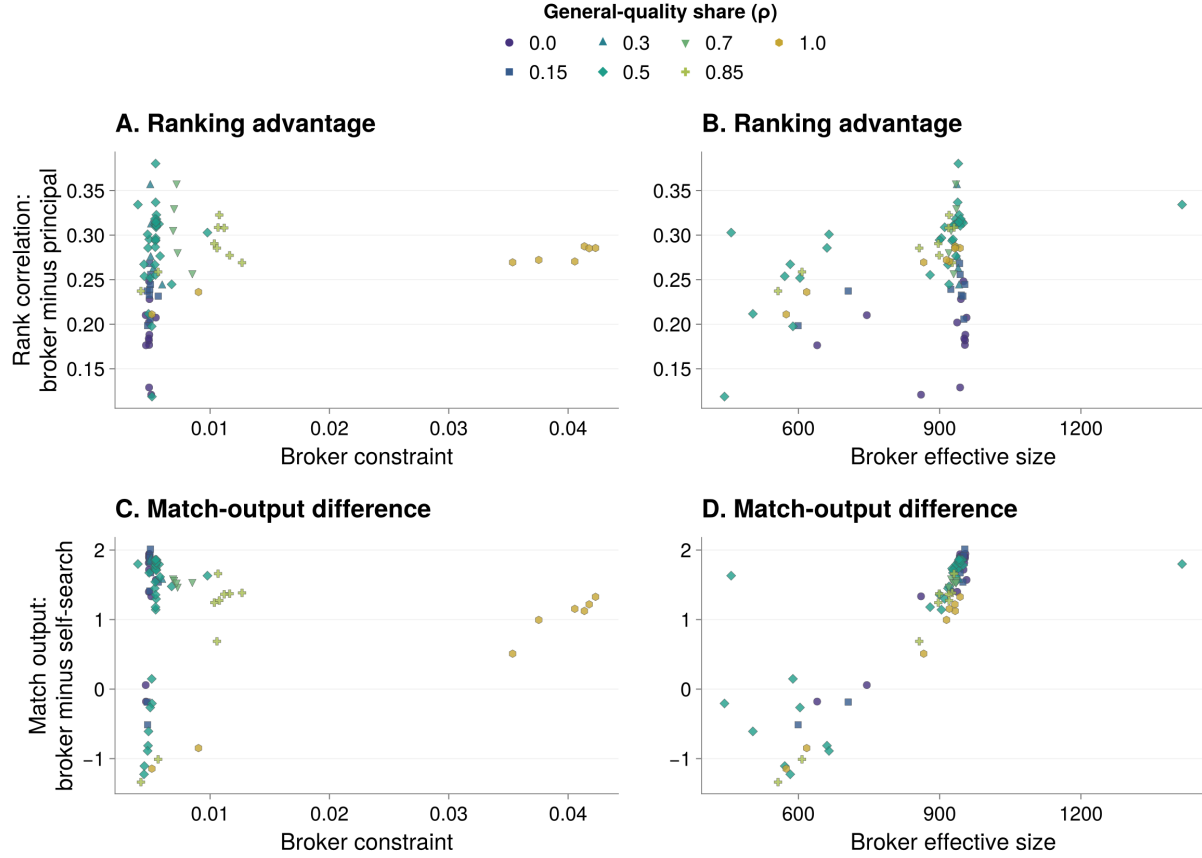
**Figure S4: Contributions of assessment and access across matching composition.** A: full service minus access-only in net output to principals per requested position. B: full service minus assessment-only. Broker models and net output are defined in the main text. Colors and symbols identify turnover  $\eta$ . Parameters other than general-quality share  $\rho$  and turnover remain at baseline. Points are paired late-window differences over 20 seeds per condition; bars are pointwise 95% Monte Carlo intervals. Positive values indicate higher net output under full service. Panels share a vertical scale; lines connect tested conditions as visual guides.



**Figure S5: Alternative structural measures across the matching problem.** A: Burt constraint, the concentration of the broker's direct and indirect ties. B: effective size, which discounts broker contacts connected to one another. Points are late-window means across the  $\rho \times \delta$  design, with other parameters at baseline. Colors and symbols identify difficulty  $\delta$ . N/A denotes the unconnected gray diamond at  $\rho = 1$ : difficulty affects only complementarity and is not applicable under pure general quality. Whiskers are 95% Monte Carlo intervals.



**Figure S6: Alternative structural measures over time and against access.** Top: the broker's Burt constraint. Bottom: the broker's effective size. A and C: baseline dynamics, recorded every 20 periods and smoothed within each seed using a 5-observation rolling mean. Lines are ensemble means; ribbons are pointwise 95% Monte Carlo intervals. Time starts at  $t = 30$ . B and D: late-window means for the 26 regimes that vary one parameter at a time from baseline. Vertical scales differ between baseline dynamics and cross-regime comparisons.



**Figure S7: Ranking and output differences by alternative structural measures.** A–B: ranking advantage, broker minus principal. C–D: mean realized output per completed match, broker minus self-search, before costs and fees. Horizontal axes show the broker’s Burt constraint (A, C) or effective size (B, D). Each point is a late-window mean for one of the 80 tested regimes. Colors and symbols identify general-quality share  $\rho$ .